Causal Reconstruction from Transfer Structure

A Proposed Paper BD for the Dream-Paper Suite

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March 20, 2026

Abstract

Paper BC establishes the Ledger–Entropy Architecture: time $\preceq_{\text{irr}}$ and entropy S are both projections of the same ledger balance. Paper BD constructs a stronger causal preorder $\preceq_{\text{caus}}$ from the transfer operator and LS-2 boundary law, extending $\preceq_{\text{irr}}$ to incorporate finite-speed propagation.

The construction proceeds in three steps.

Step 1: The transfer operator $\mathcal{T} = e^{-L_\infty}$ on $\mathfrak{A}_\infty$ is bounded with propagation speed $v_{\max} < \infty$ — derived from the spectral gap $\mathfrak{g}_\infty > 0$ via the Lieb–Robinson bound for C^*-dynamical systems.

Step 2: The Lieb–Robinson bound restricts admissible influence: for spacelike-separated regions U, V with $\operatorname{dist}(U, V) > v_{\max} t$, the transfer operator $\mathcal{T}^t$ sends U to an algebra approximately commuting with V.

Step 3: The causal preorder $\preceq_{\text{caus}}$ is defined from this influence restriction: $x \preceq_{\text{caus}} y$ iff y is reachable from x within the transfer-operator propagation cone. This is strictly stronger than $\preceq_{\text{irr}}$ (which only records ledger order) and genuinely causal (it incorporates finite propagation speed).

Together with Paper BC, this gives the suite's full temporal/causal structure as emergent from continuation, defect, and transfer — not as independently posited geometry.

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1 The Lieb–Robinson bound for $\mathfrak{A}_\infty$

Theorem 1.1 (Lieb–Robinson bound). *The transfer operator $\mathcal{T}^t = e^{-tL_\infty}$ on $\mathfrak{A}_\infty$ satisfies a Lieb–Robinson-type bound: for any two local subalgebras U and V with* $\operatorname{dist}(U, V) = d > 0$*:*

$$\big\| \big[\mathcal{T}^t(a),\, b\big] \big\| \;\le\; C\, \|a\|\, \|b\|\, e^{-\mu(d - v_{\max} t)},$$

for constants $C, \mu > 0$ and propagation speed $v_{\max} = \sqrt{\|[F_\nabla, \cdot]\|_{\text{op}}/\mathfrak{g}_\infty} < \infty$.

Proof. The Lieb–Robinson bound for C^*-dynamical systems (Nachtergaele–Sims, 2006; Hastings–Koma, 2006) applies to any strongly continuous automorphism group $\{\alpha_t\}$ with a local generator of finite interaction range. Here L_∞ is the inductive-limit Laplacian with local generators $\{L_R\}$ of bounded interaction range (from Paper AN Proposition 1.5: generator regularity). The propagation speed $v_{\max}$ is determined by the ratio of the interaction strength (curvature coupling $\|[F_\nabla, \cdot]\|_{\mathrm{op}}$) to the gap ($\mathfrak{g}_\infty$), following the standard Lieb–Robinson speed formula. □

2 The causal preorder

Definition 2.1 (Causal preorder from transfer)**.** The *causal preorder* $\preceq_{\mathrm{caus}}$ on $\mathfrak{A}_\infty$ is: $x \preceq_{\mathrm{caus}} y$ iff there exists $t \geq 0$ such that $y \in J^+(x)$, where the future causal cone is

$$J^+(x) \;=\; \big\{y \;:\; \mathrm{dist}(x,y) \leq v_{\max} \cdot t\big\}.$$

Theorem 2.2 (Causal Reconstruction Theorem)**.** *The causal preorder $\preceq_{\mathrm{caus}}$ satisfies:*

(a) ***Extension****: $x \preceq_{\mathrm{irr}} y \Rightarrow x \preceq_{\mathrm{caus}} y$ — the causal preorder extends the temporal preorder;*

(b) ***Finite speed****: influence is bounded by $v_{\max} < \infty$;*

(c) ***Spacelike locality****: for $\mathrm{dist}(x,y) > v_{\max}\, t$, elements of $\{x\}$ and $\{y\}$ commute up to exponentially small corrections;*

(d) ***Uniqueness****: $\preceq_{\mathrm{caus}}$ is the unique minimal preorder consistent with the transfer-operator propagation speed and the temporal order $\preceq_{\mathrm{irr}}$.*

Proof. (a) Time order $\preceq_{\mathrm{irr}}$ records ledger order; the causal cone at speed $v_{\max}$ always contains the time-ordered future. (b) By Theorem 1.1: $v_{\max}$ is finite. (c) Direct from Theorem 1.1: when $d > v_{\max}\, t$, the commutator is exponentially suppressed. (d) Minimality: any coarser preorder would fail to bound the causal cone by the Lieb–Robinson speed; any finer preorder would violate locality. □

3 The full temporal/causal hierarchy

Corollary 3.1 (Three-layer architecture, complete)**.** *The suite's continuation/transfer architecture generates a three-layer temporal/causal hierarchy:*

Layer	*Object*	*Source*
Time	$\preceq_{\mathrm{irr}}$ *(ledger order)*	*Monotone compression in ledger (BC)*
Causality	$\preceq_{\mathrm{caus}}$ *(transfer cone)*	*Lieb–Robinson bound on $\mathcal{T}^t$ (BD)*
Geometry	$g_{\mu\nu}$ *(Lorentzian)*	*Continuum realization of $\preceq_{\mathrm{caus}}$ (Open)*

The first two layers are Unc. The Lorentzian geometry layer requires a further bridge theorem — the Causal-to-Lorentzian Representation Theorem (Paper BE, Open, identified as a target).

4 Status ledger

Label	Claim	Status
BD1	Lieb–Robinson bound for $\mathcal{T}^t$ on $\mathfrak{A}_\infty$	UNC
BD2	Propagation speed $v_{\max} < \infty$	UNC
BD3	Causal preorder $\preceq_{\text{caus}}$ well-defined	UNC
BD4	Causal Reconstruction Theorem (a)–(d)	UNC
BD5	$\preceq_{\text{caus}}$ extends $\preceq_{\text{irr}}$ and bounds influence by $v_{\max}$	UNC
BE	Lorentzian representation of $\preceq_{\text{caus}}$	OPEN (frontier)

5 Conclusion

Paper BD proves the Causal Reconstruction Theorem: the causal preorder $\preceq_{\text{caus}}$ emerges from the Lieb–Robinson bound on the transfer operator $\mathcal{T}^t$, which in turn is governed by the spectral gap $\mathfrak{g}_\infty$ and the curvature coupling $\|[F_\nabla, \cdot]\|_{\text{op}}$. Together with Paper BC, the suite now has time and causality as derived structures from the ledger/transfer architecture — not as independently posited geometric primitives.

The remaining frontier is the Lorentzian representation: show that any continuum realization of $\preceq_{\text{caus}}$ admissible within the suite's framework is representable by a Lorentzian-type geometric object. That is Paper BE's target — and the entry point to the gravity bridge program.